

Read Section 3.2 of *Levin*.

From *Levin*, complete the following:

Section 3.2: # 1, 5, 6, 7

1. Solution is in the book.
5. Prove that for all integers n , it is the case that n is even if and only if $3n$ is even. That is, prove both implications: if n is even, then $3n$ is even, and if $3n$ is even, then n is even.

Proof. We begin by proving directly that *if n is even, then $3n$ is even*.

As n is even, then there is some integer k such that $n = 2k$, by the definition of even. Now, consider $3n$.

$$\begin{aligned} 3n &= 3(2k) \\ &= 2(3k) \end{aligned}$$

Let $\ell = 3k$ and we notice that ℓ is an integer since 3 and k are integers. Thus, we have shown $3n = 2\ell$. This meets the definition of even, and therefore, $3n$ is even.

Now we will prove by contrapositive that *if $3n$ is even, then n is even*. The contrapositive of this statement is that *if n is not even, then $3n$ is not even*. Of course, if an integer is not even, then it is odd. So what we will really prove is that *if n is odd, then $3n$ is odd*.

As n is odd, there is some integer k such that $n = 2k + 1$, by the definition of odd. Now consider $3n$.

$$\begin{aligned} 3n &= 3(2k + 1) \\ &= 6k + 3 \\ &= 6k + 2 + 1 \\ &= 2(3k + 1) + 1 \end{aligned}$$

Let $\ell = 3k + 1$ and this is an integer. Therefore, $3n = 2\ell + 1$ which is the definition of an odd integer, proving the contrapositive is true: *if n is odd, then $3n$ is odd*. This shows that the original conditional statement is true, *if $3n$ is even, then n is even*. This completes the biconditional proof of *n is even if and only if $3n$ is even* for any integer n . $\square$

6. Prove that $\sqrt{3}$ is irrational.

Before we prove this, recall the fact that *for any integer n , if n^2 is a multiple of 3, then n is a multiple of 3*.

Proof. We begin by rewriting this as the statement *if $x = \sqrt{3}$, then x is irrational*. We proceed by contradiction. So we assume that $x = \sqrt{3}$ and x is not irrational. Thus x is rational, which means that x can be written as a fraction of integers p and q . We also can make the further assumption that

p and q have no common factors. Therefore, we have

$$\begin{aligned}\sqrt{3} &= \frac{p}{q} \\ (\sqrt{3})^2 &= \left(\frac{p}{q}\right)^2 \\ 3 &= \frac{p^2}{q^2} \\ 3q^2 &= p^2.\end{aligned}$$

This tells us that p^2 is a multiple of 3. By the fact before the proof, this tells us that p is a multiple of 3. And therefore, there is some integer k such that $p = 3k$. Squaring this equation, we get that $p^2 = 9k^2$. Then, we substitute this into the main equation we are working with and get

$$\begin{aligned}3q^2 &= p^2 \\ 3q^2 &= 9k^2 \\ q^2 &= 3k^2.\end{aligned}$$

Then we see that q^2 is a multiple of 3 and again, by the fact before the proof, q is a multiple of 3. But we assumed at the beginning that p and q have no common factors and we just found that they both must be multiples of 3, which is a contradiction. Thus our assumption that x is rational must be false and therefore we have shown that if $x = \sqrt{3}$, then x is irrational. $\square$

7. Consider the statement: for all integers a and b , if a is even and b is a multiple of 3, then ab is a multiple of 6.

(a) Prove the statement. What sort of proof are you using?

Proof. We prove this directly. Since a is even, then there is some $k \in \mathbb{Z}$ such that $a = 2k$. Since b is a multiple of 3, then there is some $j \in \mathbb{Z}$ such that $b = 3j$. Then we have

$$\begin{aligned}ab &= (2k)(3j) \\ &= 6kj\end{aligned}$$

Let $\ell = kj$ and therefore $ab = 6\ell$. This shows that ab is a multiple of 6. $\square$

(b) State the converse. Is it true? Prove or disprove.

The converse of the statement $P \rightarrow Q$ is the statement $Q \rightarrow P$. So our converse is *for all integers a and b , if ab is a multiple of 6, then a is even and b is a multiple of 3*.

Counterexample. We need this to be true for all integers that multiply together to yield a multiple of 6. In particular, if $ab = 6$, the possible factors, ignoring order, are $(1, 6)$ and $(2, 3)$. The pair $(1, 6)$ does not satisfy this, since 1 is not even, nor is it a multiple of 3. But $1 \cdot 6 = 6$. Thus, we have found a counter example to the converse statement and this shows it is false.